

01 | Day 4: Think Like an ML Engineer

The model trained. Is it good, and what should improve next?

Retrieve: your original Day 3 live response and appended consolidation in **Day 3 Checks**.

02 | Which Link Fails First?

Compare three anonymized Day 3 consolidation chains:

1. bounded claim + real limitation + discriminating experiment;
2. limitation that cannot threaten the claim;
3. next experiment that changes too many things.

Revise only the failed link in your own chain.

03 | Today's Evidence Arc

consequence -> metric/threshold -> curves -> slices -> one experiment -> reproducibility -> resources -> transfer -> defense

At every step:

claim -> evidence -> alternative -> decision -> next discriminating check

450 minutes: protected breaks, lunch, and final buffer

04 | Start with the Consequence

Before a metric, ask:

1. What decision will be made?
2. Which class is positive?
3. Who experiences false positives and false negatives?
4. What evidence would change the operating point?

Accuracy can be correct and still answer the wrong question.

05 | Confusion Matrix = Consequence Map

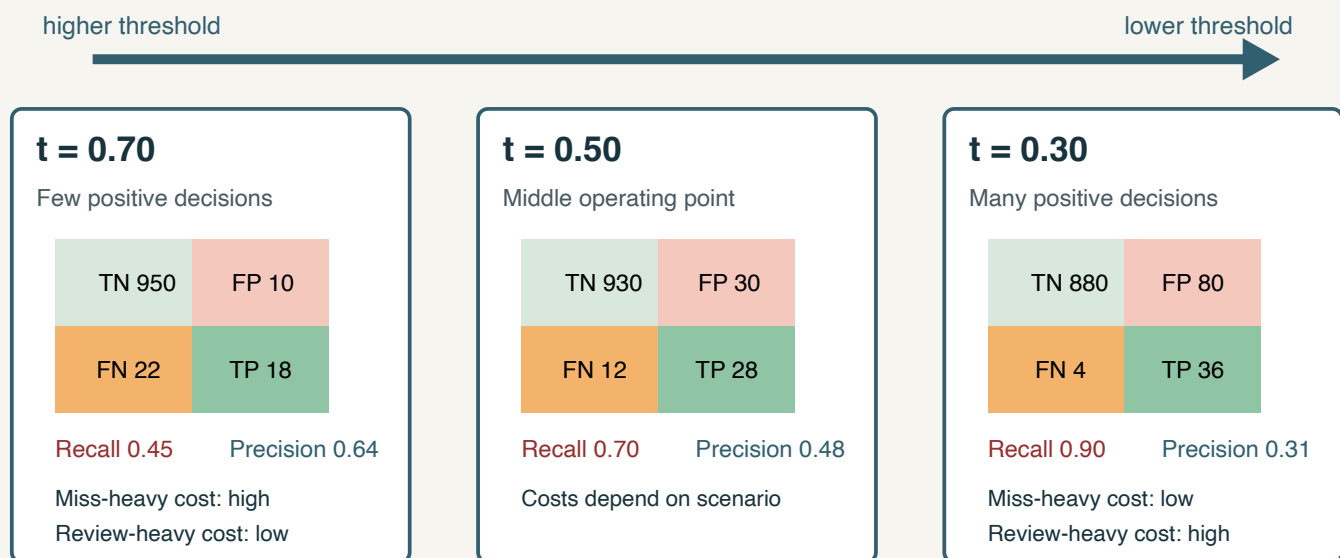
	Predicted negative	Predicted positive
Actual negative	TN	FP
Actual positive	FN	TP

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}$$

Rows are true labels; columns are predicted labels.

06 | Move the Threshold, Move the Decision

Same scores, different operating decisions



Moving the threshold changes decisions, not learned parameters or score ranking.

07 | ACT-D4-01: Cost Council

Scenario A:

$$\text{cost} = 5(FP) + 250(FN)$$

Commit before the reveal:

- threshold direction;
- expected benefit and harm;
- minimum metric/count set;
- evidence that would reverse the decision.

Then the cost ratio changes.

08 | F1 Is a Choice, Not a Universal Answer

$$F_1 = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

F1 weights precision and recall equally.

It does **not** encode:

- a 50:1 error cost;
- review capacity;
- delayed or uneven harm;
- stakeholder policy.

09 | PR and ROC: Overview, Not an Operating Answer

- Precision-recall: how positive-decision quality and coverage trade over thresholds.
- ROC: true-positive rate versus false-positive rate over thresholds.
- ROC AUC: ranking summary across thresholds.

AUC does not choose a deployment threshold.

The operating point still needs consequences and constraints.

10 | LAB-D4-01: Accuracy Is Not Enough

Predict: majority-baseline minority recall and threshold direction.

Observe: accuracy beside recall/F1, confusion counts, PR point, and cost.

Decide: an operating point for each scenario.

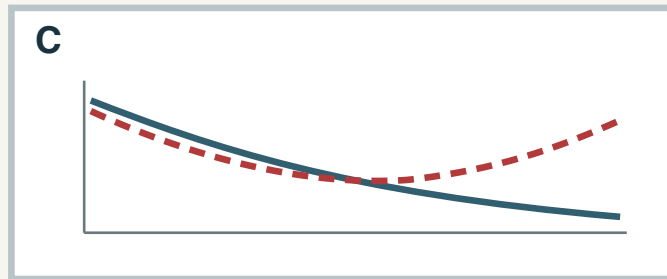
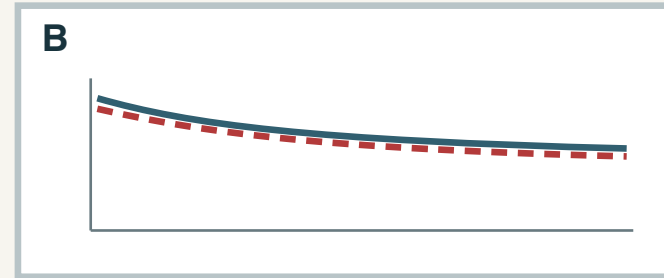
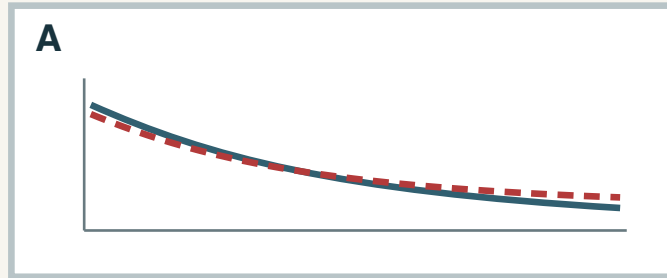
Debrief: what changed when the threshold moved, and what did not?

`courseware/day-4/labs/LAB-D4-01-accuracy-is-not-enough.ipynb`

11 | ACT-D4-02: Mystery Curves

What story fits each path?

— train - - - validation



Aligned axes. Curves support hypotheses; they do not uniquely prove causes.

12 | Diagnose the Relationship, Not One Number

Evidence	Leading family	Do not conclude yet
train + validation weak, smooth plateau	high bias / underfit	exact cause
strong train, validation worsens	high variance / overfit	split is valid
oscillation/divergence	optimization or implementation	more capacity helps
stable improvement, modest gap	healthy relative to current evidence	every slice is healthy

A small gap can still mean both are bad.

13 | LAB-D4-02: Model Detective

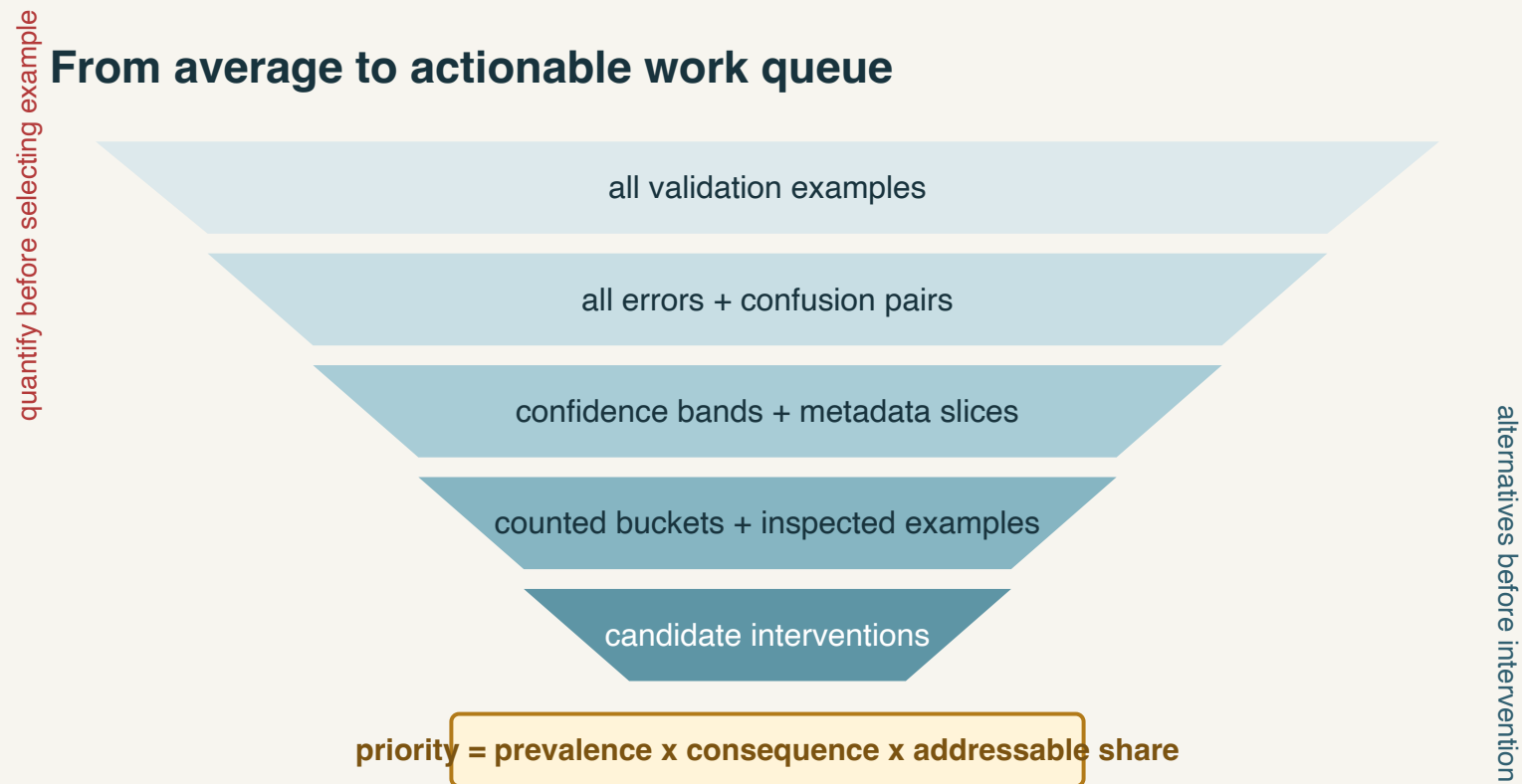
Curves -> confusion pairs -> confidence -> slices -> selected errors -> one next experiment

Before the configuration reveal:

- diagnose from two observations;
- name an alternative;
- request one discriminating evidence card.

```
courseware/day-4/labs/LAB-D4-02-model-detective.ipynb
```

14 | Error Analysis Is a Funnel



15 | Largest Bucket Is Not Automatically First

Bucket	Count	Cost	Addressable share
faint digits	24	3	0.50
3 -> 8	31	1	0.60
suspected labels	8	8	0.75

Choose a priority rule. Then state what would falsify the intervention mechanism.

16 | Before Lunch: Individual Evidence

CHECK-D4-01 | 5-minute live minimum

Accuracy risk -> threshold benefit/harm -> AUC correction + reversing assumption.

CHECK-D4-02 | 5-minute live minimum

Curve diagnosis + alternative -> quantified slice priority -> action + rejection evidence.

Append marked consolidation in Day 4 Checks by 16:30.

17 | One Experiment Should Teach One Thing

One run should answer one clear question



A clean negative result can strengthen the decision trail

It rejects or narrows a mechanism while controls remain fixed.

Confounded run

wider model + new optimizer + longer training

A score change cannot identify which factor or interaction caused it.

Write the rejection observation and stop rule before compute.

18 | A Negative Result Can Be Excellent

A strong negative result:

- preserves the baseline and controls;
- tests a plausible mechanism;
- observes the declared rejection evidence;
- removes or narrows an explanation;
- chooses a sharper next experiment.

Information gained is an outcome.

19 | Reproducibility Has a Boundary

Same seed is not the same experiment record

Incomplete record

- [x] architecture
- [x] seed = 17
- [] split hash / transforms
- [] packages / device / threads
- [] metric averaging / threshold
- [] checkpoint rule
- [] raw repeat + tolerance
- [] timing warm-up / repeats

Usable team record

- [x] code + architecture/config
- [x] data / split / transforms
- [x] RNGs + environment
- [x] metric / threshold / slices
- [x] checkpoint selection rule
- [x] raw repeats + tolerance
- [x] timing method / workload
- [x] result + limitation + next

Claim: repeatable within this environment and tolerance, not identical everywhere.

20 | Timing Is an Experiment Too

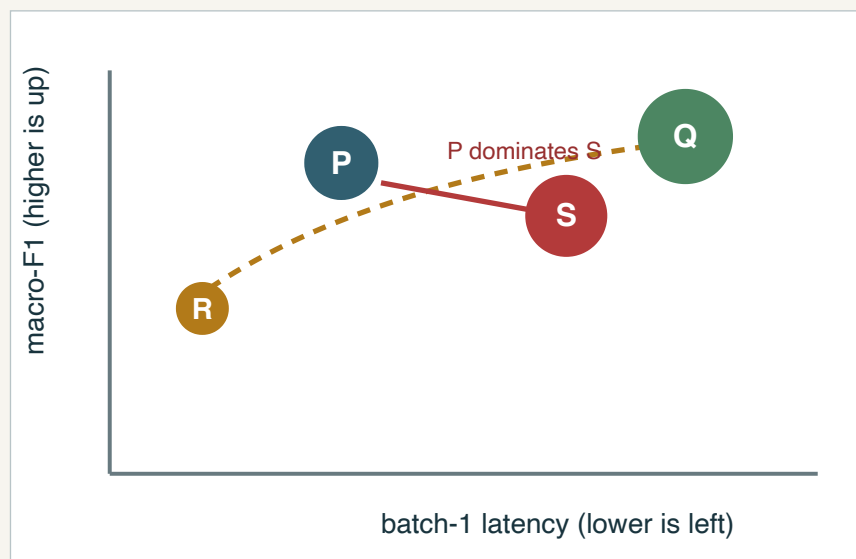
Record:

- hardware, software, threads/device;
- input shape and batch/load;
- warm-up and repeats;
- median and useful tail statistic;
- synchronization for asynchronous accelerator work.

GPU available != GPU utilized != lower end-to-end latency.

21 | Quality Lives on a Resource Frontier

There is no winner without a constraint



Interactive

latency < 6 ms -> P

Nightly batch

highest throughput -> Q

Edge device

size < 2 MB -> R

Circle size encodes model size. Hardware, load, tail latency, reliability, and energy remain unmeasured.

22 | ACT-D4-03: Constraint Changes the Winner

Round 1: batch-1 latency `< 6 ms`

Round 2: highest nightly throughput

Round 3: model size `< 2 MB`, macro-F1 `>= 0.920`

For each:

choice + evidence + concession + missing measurement

23 | LAB-D4-03: You Get One Experiment

Before: hypothesis, one major change, expected/rejecting evidence, stop rule.

During: fixed split/seed policy/metric; bounded run.

After: complete record, repeated result, quality/resource comparison, next experiment not run.

```
courseware/day-4/labs/LAB-D4-03-one-experiment.ipynb
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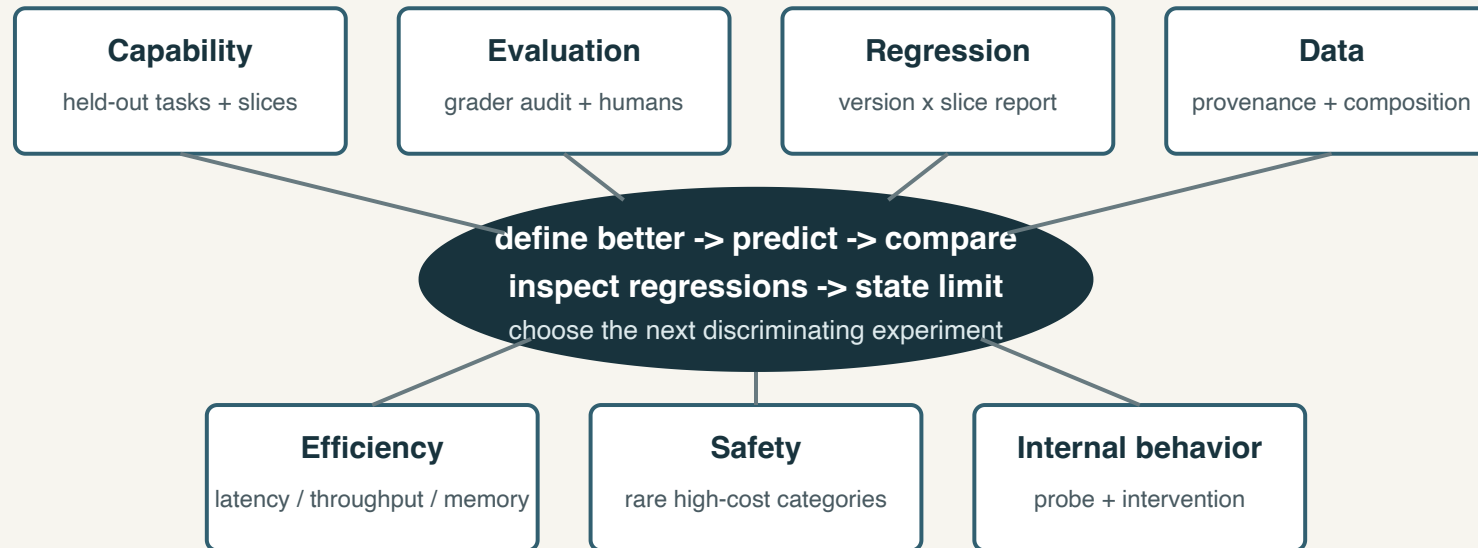
24 | Transfer the Loop, Not the Scale Claim

For a new problem:

1. define "better";
2. choose held-out evidence and slices;
3. predict what should change;
4. inspect regressions and alternatives;
5. price resources;
6. state the claim boundary;
7. choose the next experiment.

25 | Seven Representative Case Files

Different case files, same evidence discipline



Representative public categories, not a complete role map or private workflow claim.

26 | ACT-D4-04: Is It Better at Coding?

Design evidence for:

- held-out tasks and meaningful slices;
- executable grader **and its exploit path**;
- blinded human sample **and disagreement rule**;
- contamination control;
- unacceptable regression gate.

Then respond to an adversarial reveal.

27 | Scope Boundaries Protect the Investigation

SHORTEN ROC mechanics and extended taxonomy

MOVE / reference

- mixed-precision implementation;
- quantization;
- model-parallel/distributed implementation;
- deep interpretability methods;
- complete safety engineering systems.

Representative category != complete practice.

28 | Capstone Brief: Routing-Code Digits

You inherit:

- a fixed train/validation/test manifest;
- one anonymous case **A** , **B** , or **C** ;
- baseline curves, metrics, errors, and resource evidence;
- a fixed run budget;
- one final test authorization.

Small digits make evidence visible; they do not create production realism.

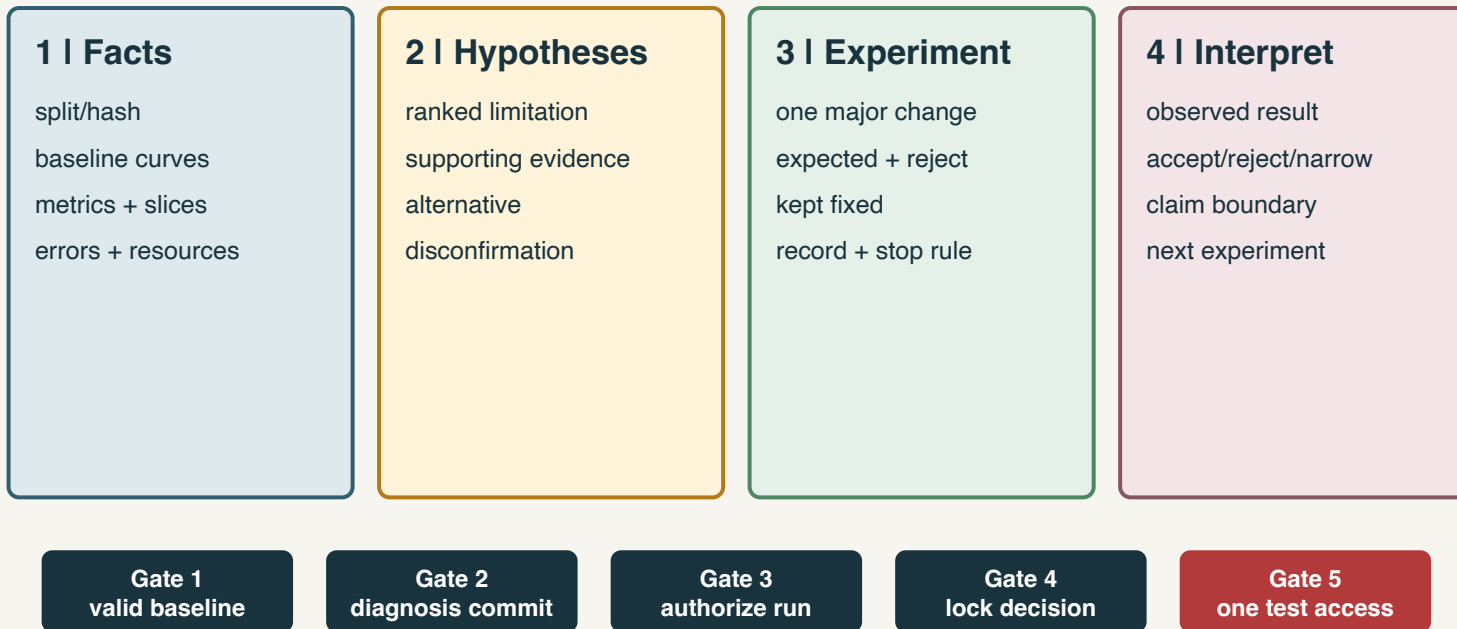
29 | Capstone Contract

- one primary experiment;
- one major factor changed;
- evidence and rejection criterion before compute;
- test locked until the decision is fixed;
- optional follow-up only if authorized;
- cached outputs visibly labeled;
- negative results preserved.

Do not spend interpretation time buying another run.

30 | The Evidence Board Has Four Lanes

Evidence board: never let facts and guesses merge



Case letters A/B/C reveal no diagnosis. Cached output must remain visibly labeled.

31 | Anonymous Means Anonymous

Your case is only:

A | **B** | **C**

Do not infer meaning from:

- letter;
- color;
- filename;
- variable name;
- another team's intervention.

Score the evidence chain, not hidden-answer matching.

32 | Capstone Rubric: 100 Points

The highest score is not automatically the strongest work

Performance 25	Generalization 20	Experiment design 20	Diagnosis 15	Efficiency 10	Explanation 10
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A controlled negative result can earn full reasoning credit.

Minute 1 Baseline wrong or uncertain	Minute 2 Evidence support + threat	Minute 3 Experiment change + prediction	Minute 4 Result quality + resources	Minute 5 Decision why + next + limit
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Team score and individual check are separate.
Test access remains locked until the decision is fixed.

33 | Five-Minute Defense

1. What was wrong or uncertain?
2. What evidence supports **and threatens** the diagnosis?
3. What one factor changed, and what was predicted?
4. What happened to quality, generalization, slices, and resources?
5. Why, what can you claim, and what comes next?

Peers submit one skeptical question.

34 | Finish Individually, Leave with the Loop

CHECK-D4-03 | 5-minute live minimum

Two inference blockers -> one-change expected/rejecting evidence -> calibrated quality claim.

CHECK-D4-04 | 5-minute live minimum

Bounded claim -> two alternatives -> decision -> supporting/rejecting next evaluation -> individual transfer.

Final loop: consequence -> evidence -> alternative -> one experiment -> interpretation -> resource cost -> honest claim